



A Comparative Analysis of Total Cost of Ownership Between GMS and Non-GMS Enterprise Devices

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Executive Summary

This whitepaper aims to explore and highlight the key distinctions between Android Open-Source Project (AOSP) and Google Mobile Services (GMS) devices. The primary focus of exploration is on their impact on the total cost of ownership (TCO) in an enterprise setting across key aspects, including security, flexibility, productivity, sustainability, & the ecosystem.

While AOSP devices offer a baseline open-source platform, GMS devices may provide a more comprehensive and user-friendly experience through the integration of Google's widely used applications. The inclusion of GMS in enterprise devices can positively influence productivity, collaboration, and overall user satisfaction.

The whitepaper concludes by establishing a comprehensive framework for TCO calculation and facilitating a comparative analysis of AOSP and GMS devices, factoring in the various costs involved throughout the device's lifetime.

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Introduction to AOSP & GMS

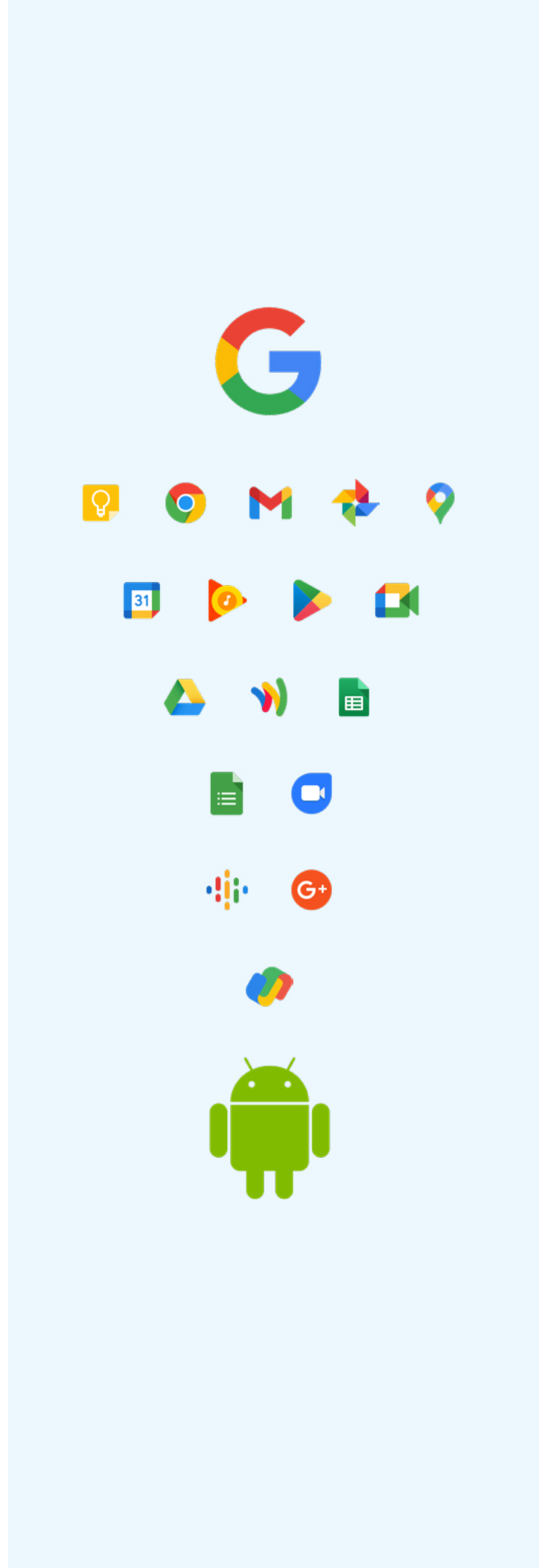
AOSP (Android Open-Source Project) is the open-source software stack for Android maintained by Google. AOSP provides standard, device-level functionalities such as email and calling. It is created for a wide array of devices with different form factors. Android's primary purpose is to create an open software platform available for carriers, OEMs, and developers to make their innovative ideas a reality and to introduce a successful, real-world product that improves the mobile experience for users.

Google Mobile Services (GMS) constitutes a compilation of Google applications and services (APIs) designed to bolster functionalities across various devices. These applications collaborate seamlessly to guarantee an optimal user experience right from the start. GMS is exclusively accessible through a licensing agreement with Google, offering a comprehensive suite of popular apps and cloud-based services. Included in this package are the Google Play Store, Google Chrome, Google Search, Google Maps, Gmail, YouTube, Google Drive, Google Photos, Google Duo, Google Meet, Google TV, Google YouTube Music, and Google Calendar. GMS services include AndroidAuto, CarrierServices, GooglePartnerSetup, GooglePackageInstaller, AndroidPlatformServices, GooglePrintRecommendationService, GoogleServicesFramework, SpeechServicesByGoogle, GmsCore, FindMyDevice, ConfigUpdater, SetupWizard, GoogleActionsService, and GoogleOneTimeInitializer. It is important to note that GMS is distinct from the AOSP stack.



Manager, Tier-1 North American OEM

The Sasken team is very skilled and has brought high value to our mobile computer projects. The development and test support we have experienced have been tremendous. When faced with new challenges, the Sasken team did everything necessary to meet our expectations.



AOSP ecosystem, what AOSP offers and significant limitations of using only AOSP

AOSP serves as the foundation for Android devices. It includes the core components of Android, such as the kernel, libraries, and system apps, but does not include any proprietary Google apps or services.

The below picture shows the AOSP Software Stack Architecture.

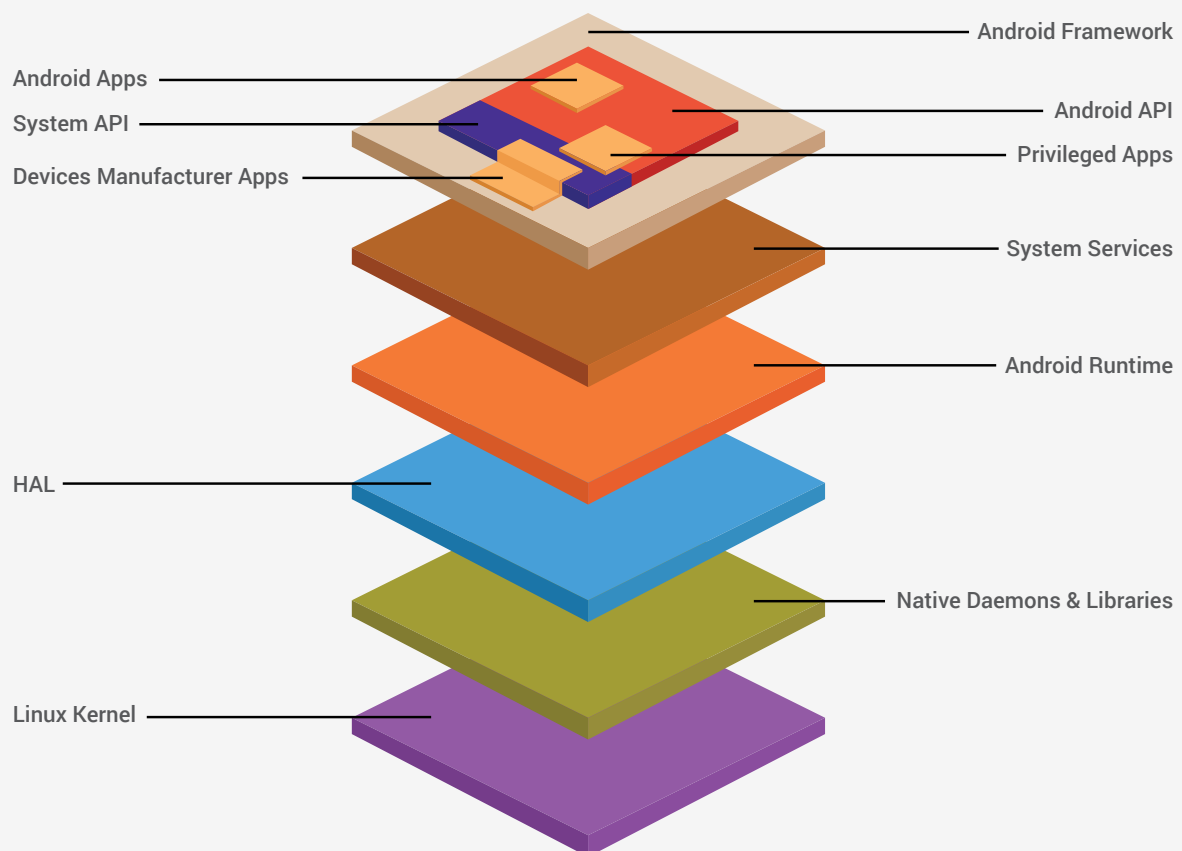


Figure: AOSP software stack architecture.

AOSP Ecosystem

Original Device Manufacturers (ODMs) and Original Equipment Manufacturers (OEMs) have a significant role in developing and manufacturing a wide range of Android-based products, including smartphones, tablets, smart TVs, smartwatches, and more. They are known for their expertise in customizing AOSP to create tailored user experiences for their brands around the world. In addition to traditional Android devices, some ODMs/OEMs have contributed to producing AOSP-based products for global giants.

AOSP for specialized use cases

AOSP is available for download, and individuals have the freedom to customize it for their devices. It offers a comprehensive and fully operational implementation feasibility of the Android mobile platform.

AOSP (or non-Google Mobile Services) may be an ideal solution for various specialized use cases in enterprises. For instance, it is well-suited for self-service kiosks and information stations in environments like museums, airports, and libraries, where tailored software experiences are key.

Additionally, AOSP devices could be suitable for medical and healthcare devices, including patient monitoring equipment, diagnostic tools, and specialized medical apps. Additionally, digital signage solutions used extensively for displaying advertisements, information, or interactive content may benefit from leveraging AOSP devices.

Limitations of AOSP

While AOSP offers flexibility, it may lack support for apps like cloud messaging or an advanced location services app. It might contain a limited selection of end-user apps, potentially necessary for specific device types.

AOSP is distinct from what most people commonly associate with Android due to its notable absence of key elements such as the Google Play Store, Google Play Services, and most Google apps. AOSP is almost unrecognizable as a Google platform for the end user.

Consequently, AOSP, in its standard form, may need more time to be ready for use as a complete and functional operating system after installation for specific enterprise use cases.

Some of the key features enabled by Google Play services include:

- **Google Maps Integration:** Many apps and services rely on Google Maps for location data, mapping, and navigation. Without Google Play services, these functionalities may be limited or non-functional.
- **Google Drive Integration:** Apps like Google Drive and Google Photos use Google Play services for seamless cloud storage and synchronizing. AOSP devices may not have direct integration with these services. However, using the Google Drive API, you can integrate Drive directly into your Android app without any reliance on GMS.
- **Google Assistant:** Google Assistant's advanced features and voice recognition capabilities are tightly integrated with Google Play services. Without them, you may have limited access to these features.
- **SafetyNet:** SafetyNet is a Google Play service that many apps use to check the security and integrity of the device. Some apps, especially banking and payment apps, may only work with this service.
- **Firestore Cloud Messaging (FCM):** Many apps, including popular messaging and notification apps, rely on FCM for push notifications. AOSP devices may receive push notifications less efficiently.
- **Automatic App Updates:** Google Play services facilitate automatic app updates, ensuring apps are up to date without user intervention. AOSP devices might not fully support automatic app updates.
- **Backup and Restore:** AOSP devices may not have access to Google's backup and restore features, which can impact app data backup and device setup.
- **Location Services:** While AOSP provides basic location services, Google Play services offer more advanced and accurate location data, which many apps depend on.

Some of the key features of Android, like translating text from photos using the camera, transcribing spoken words, the live captions feature, and the smart reply feature in the Messages app, might not work out of the box.

Some AOSP-based Android distributions have Aurora Store (formerly Yalp Store), F-Droid, Aurora Droid, and Aptoide to emulate the Google Play Store. The installed applications might not work or have limited functionality because they depend on only APIs provided through Google Play Services.

Several key features and functionalities in Android depend on Google Play services and may not work properly with a pure AOSP build.



GMS

Android integrates cutting-edge security features and collaborates with developers and device implementers to ensure the safety of the Android platform and its ecosystem. A strong security model is crucial for fostering a dynamic ecosystem of apps and devices centered on the Android platform and backed by cloud services. Consequently, Android undergoes a thorough security program throughout its entire development lifecycle.

Google provides a set of cloud-based services available for compatible Android devices with Google Mobile Services. These services are not part of the AOSP.

The primary Google security services are:

- **Google Play:** Google Play serves as a set of services enabling users to explore, install, and buy apps either from their Android device or through the web. It simplifies the process for developers to connect with Android users and prospective customers. Additionally, Google Play offers features such as community reviews, app license verification, security scanning, and other security services.
- **Android updates:** The Android update service provides selected Android devices with new features and security updates, delivering them either through the web or via over-the-air (OTA) updates.
- **App services:** Frameworks enabling Android applications to leverage cloud capabilities, including tasks like backing up app data and settings, as well as implementing cloud-to-device messaging (C2DM) for push messaging.
- **Verify Apps:** Alert or automatically prevent the installation of malicious apps and consistently scan apps on the device, providing warnings or removing harmful apps as needed.
- **SafetyNet:** A privacy-preserving intrusion detection system to assist Google in tracking and mitigating known security threats and identifying new ones.
- **SafetyNet Attestation:** An anti-abuse API that allows app developers to assess the Android device their app is running on. The SafetyNet Attestation API provides a cryptographically-signed attestation, assessing the device's integrity. Google plans to turn down the SafetyNet Attestation API starting January 2024 and replace it with the [Play Integrity API](#), consolidating multiple integrity offerings (including the SafetyNet Attestation integrity verdict) under a single API.
- **Android Device Manager:** A web application and an Android application designed for locating a lost or stolen device.

While GMS offers productivity applications and services, if a user does not use them, they can take up unnecessary space on the device. ODM/OEMs also need to undergo the GMS certification cycle with every release of software.

MADA, EDLA & steps for getting license

MADA (Mobile Application Distribution Agreement) is the official contract through which a GMS license is granted to OEMs/ODMs (brands/manufacturers) manufacturing the device. The license allows OEMs to pre-install Google's suite of apps and services on their Android devices and offer a full Android experience with Google's apps and services integrated into the device. The license is non-transferable.

The EDLA, known as the Enterprise Device License Agreement, parallels the GMS license but is tailored for OEMs and ODMs focused on manufacturing devices for enterprise use. Just as the GMS license facilitates the inclusion of Google's apps and services, the EDLA grants enterprise-focused device manufacturers the capability to pre-install these essential Google services on their devices, providing enterprise users access to a comprehensive suite of productivity and connectivity tools.

Google is introducing the Enterprise Device License Agreement (EDLA) with select changes from MADA, including:

- Extended launch & software approval windows.
- Screen sizes up to 70" & battery-less devices are allowed.
- Introduction of a license fee to support the program.
- Support extended security patch support from device launch & upgrades.

Select form factors such as point-of-sale (PoS) devices, kiosks, digital displays, and "headless" 2-piece units that were previously ineligible under the MADA may be approved for GMS under the EDLA.

Getting MADA license

- Contact Google by filling out the below form. https://www.android.com/intl/en_in/gms/contact/
- Google will assess your eligibility for a GMS license. This involves evaluating your company, business plan, and commitment to adhering to Google's requirements and policies. Google typically works with established device manufacturers (ODMs) and original equipment manufacturers (OEMs).



- Once Google determines that you meet the eligibility criteria, you will agree with Google to discuss the terms of the MADA, including licensing fees, terms and conditions, and any other business arrangements.
- After a successful discussion and review by Google, the company gets MADA- the GMS license.

Getting EDLA License

- Get the EDLA form, available for review from the OEM/ODM's business development contact.
- 3PL: EDLA partners will use the same 3PL partners that have been managing approvals under MADA.
- OEM/ODM partner has to sign the EDLA agreement.

For further information, please contact
marketing@sasken.com

Unlocking the Power of Google Mobile Services (GMS)

Google Mobile Services (GMS) offers an extensive suite of applications and services that may benefit Android device users and manufacturers.

Together, these applications enhance the user experience by providing powerful tools for communication, productivity, entertainment, and navigation. The Google Play Store expands this experience by connecting users to a vast ecosystem of apps. At the same time, Google Play Services ensures the seamless operation of a diverse array of Android applications. This comprehensive suite significantly elevates user satisfaction and the overall value of Android devices.

For device manufacturers, pre-installing GMS apps may equip their products with a well-rounded package of essential, high-quality applications, bolstering user satisfaction and amplifying the desirability and security of their devices. Additionally, access to the Google Play Store is a gateway to a broad spectrum of apps, rendering the device more versatile and capable of accommodating users' diverse preferences.

Furthermore, integrating enterprise-focused tools such as Android Enterprise, zero-touch enrollment, and robust device management solutions into the GMS framework allows the business sector to manage Android devices securely and efficiently within an enterprise environment. The overall consistency and integration of GMS apps within the Android ecosystem guarantee a seamless and user-friendly experience, cementing GMS as a valuable and indispensable component contributing to the success of Android-based devices in the competitive marketplace.

Security on GMS devices

Google Play System Updates

Google has introduced Project Mainline to offer increased control and flexibility when delivering critical updates to devices with Google Mobile Services (GMS). In addition to the monthly Android Security Bulletin (ASB), which addresses potential issues impacting Android devices, Project Mainline provides a means to enhance the security and delivery of important updates to GMS devices.

From Android 10 onwards, Google introduced a feature called "Google Play System Updates," which enables Google to deliver critical software updates to devices with Google Mobile Services (GMS) directly. This means that you do not have to wait for your device manufacturer to release an update. These updates target various core components of the Android operating system and help enhance security, performance, and functionality.

Initially, in Android 10, there were thirteen mainline modules included in this update mechanism. Mainline modules are crucial parts of the Android system that can be updated by Google independently from complete OS updates. This allows for faster delivery of important improvements.

With the release of Android 11, the number of [Mainline modules](#) increased to 25. This expansion represents a broader range of system components that can receive timely updates. It is a significant enhancement in Android's approach to addressing security issues and ensuring that more devices can benefit from these improvements without relying solely on device manufacturers to roll out updates.

Google Play System updates are delivered as an [APEX package](#) that Google has specifically developed. Some of the modules are also provided as an APK container.

Disabling 2G Support & Null-ciphered cellular connectivity

Android 12 introduced the capability to deactivate 2G support at the modem level, thus safeguarding users from the inherent security risks associated with 2G's outdated security model.

Recognizing the crucial importance of turning off 2G for enterprise customers, Android 14 extends this security feature to Android Enterprise. It introduces the ability for IT administrators to restrict managed devices from downgrading to 2G connectivity and thus from potential security threats.



Android 14 also introduces a feature that deactivates support for null-ciphered cellular connectivity. Google ensures the full integration of these features and provides a user-friendly interface, such as a toggle option for turning off 2G support. This ensures that these security measures work seamlessly out of the box.

However, it is important to note that in the case of AOSP, there is no guarantee that these features will be incorporated, as it is up to the device manufacturers or ODMs to make the above features functional.

SafetyNet for apps

SafetyNet is a feature provided by Google Mobile Services (GMS) that helps ensure the security and integrity of a device and the apps running on it. It is primarily used to protect against tampering, unauthorized access, and the abuse of certain functionalities, and it is essential for apps that require a high level of security, such as mobile banking apps, payment apps, and some gaming applications.

SafetyNet enhances the security of GMS devices and the apps they run. Still, it can also limit the functionality of devices rooted or modified in ways considered insecure. App developers can integrate SafetyNet checks into their apps to ensure they run on secure, unaltered devices. This is crucial for apps that manage sensitive user data or transactions.

SafetyNet can verify whether a device has hardware-backed security features, such as a Trusted Platform Module (TPM) or a secure element, which is important for securely storing encryption keys.

Rich Suite of Enterprise Security Features

Market research firm Omdia highly rates Android GMS devices for their robust enterprise security features. These devices offer extensive capabilities, including jailbreak/root detection, certificate management, device management, app management, and various management modes such as bring your own device (BYOD) and company-owned options. Additionally, GMS devices support enterprise app distribution, allow for efficient enterprise wipe procedures, provide enterprise-level access features such as VPN, support enterprise authentication methods, and enable remote audit logging. This comprehensive suite of security and management features makes Android GMS devices a preferred choice for organizations seeking reliable and secure mobile solutions for their enterprise needs.

Flexibility, Productivity & Sustainability on GMS devices

Using a single mobile device for both work and personal use has its advantages but also presents unique challenges. One of the critical concerns is the need to maintain the security of company data while respecting personal data privacy. Android's work profile feature, managed through Managed Google Play, offers an elegant solution to this dilemma. The work profile protects company data from personal apps and data. Individual apps and data are kept private from IT administrators, even if the company owns the device. At the end of the workday, employees can pause their work profile and seamlessly transition to using their personal apps, ensuring a clear boundary between their professional and personal lives. This eliminates the need to carry multiple phones and contributes to reducing electronic waste, making it a practical and eco-conscious solution for modern work-life balance.

Android Enterprise Recommended is committed to delivering dependable, sustainable, and innovative hardware solutions from manufacturers, prioritizing extended software support. These GMS devices are designed to cater to diverse use cases, spanning smartphones, tablets, dedicated devices, rugged equipment, kiosks, point-of-sale systems, digital signage, etc. Their durability and reparability set these devices apart, ensuring longevity and reducing electronic waste. The Android Enterprise Recommended program offers a seal of quality for businesses and organizations seeking dependable hardware that can thrive across a broad spectrum of professional applications while benefiting from enduring software support.

Zero-touch enablement for Android Enterprise is a sophisticated method for efficiently provisioning and configuring Android devices in an enterprise or corporate context. A notable feature of this process is its seamless integration with Google Mobile Services (GMS), which significantly enhances the overall experience.



North American OEM

As a team, Sasken has demonstrated great dedication to our engagement. Working with Sasken gives me confidence that they will go the extra mile when required. The team leaders are encouraging of the team. It is evident from their good work that the entire team is positively engaged throughout the duration of our engagement.



How Zero-touch enablement works

Automatic Device Enrollment: Zero-touch enrollment ensures that Android devices equipped with Google Mobile Services are automatically enrolled in an enterprise's mobile device management (MDM) system as soon as they are powered on and connected to the internet. This eliminates the need for manual setup by IT administrators or end-users.

Pre-configured Profiles: Administrators can effortlessly pre-configure device settings, policies, and apps through their MDM solution, all while leveraging the capabilities of GMS. When enrolled, a device automatically receives and applies these pre-configured profiles.

User-Friendly Experience: Zero-touch enrollment provides a seamless and user-friendly experience. End-users simply turn on the device and connect it to the internet, and the device is automatically set up with the necessary configurations, apps, and security policies.

Security and Management: The automatic enrollment ensures that devices are secure and compliant with organizational policies from the moment they are activated. Thanks to the robust set of GMS services, IT administrators can remotely manage and monitor the devices.

Efficiency: Zero-touch enrollment significantly reduces the time and effort required for device provisioning. This is especially valuable for large-scale deployments in businesses and organizations.

Device Choice: Zero-touch enrollment supports various Android devices, making it suitable for multiple use cases and device types. It provides a consistent and reliable experience across a diverse device ecosystem.

In partnership with GMS, zero-touch enablement not only simplifies the process of provisioning and managing Android devices for business purposes but also enhances security, efficiency, and ease of use for IT administrators and end-users. This approach streamlines the deployment of Android devices and underscores the benefits of integrating GMS services in an enterprise environment.

GMS Express program and how it helps OEMs

The new Android conformance and compatibility program (started in 2017) will help smartphone manufacturers build devices using SoCs with pre-installed Google's Mobile Services (GMS).

What GMS Express offers

Previously, SoC vendors provided their partners with low-level support through a Board Support Package (BSP), hardware-specific drivers, and a version of Android based on AOSP.

GMS Express lets SoC vendors ship pre-certified and CTS-compliant Android builds with Google Mobile Services that are ready for Google's Compatibility Test Suite (CTS) certification by third-party labs, ensuring quality Android experiences and faster go-to-market.

"OEM partners can potentially reduce the typical compatibility certification process timeline from three months to four weeks."



Manager, Tier-1 Japanese OEM

Sasken has been collaborating with us in the areas of end-to-end support for phone/field testing for Sony handsets. Their capability of delving into all aspects of testing, including new features like VoLTE, voice over Wi-Fi, carrier aggregation, and complying with the testing regulations of operators, was commendable. They also helped Sony take measures to check the quality and performance of their handsets, which made the commercial launch of the handset smoother across the AMEA region. We are extremely satisfied with their well-timed support, expertise, and skill set that they put forth.



Calculating TCO for a GMS and non-GMS Enterprise device

In this section, we will define what TCO (total cost of ownership) means for an enterprise device or any asset and develop a framework to calculate TCO, followed by a TCO comparison between a GMS and a non-GMS enterprise device.

What is TCO?

TCO, or total cost of ownership, measures all the costs associated with acquiring, using, and maintaining an asset or device over its lifetime. In other words, it does not only include the initial purchase price but also the operating costs, maintenance costs, training costs, support costs, and any potential risks or hidden costs. It is a measure used by enterprise customers to compare different options and make informed decisions to reduce IT costs and improve operational efficiencies.

How to calculate TCO for an Asset?

Although there have been many analyses to determine the TCO for an asset or device over the years, there is no standard way to calculate this. We could develop a framework to evaluate the TCO for an asset or device like the one provided below:

- Define the scope and objectives of the analysis after identifying the asset or device for which TCO needs to be evaluated.
- Identify and categorize all the costs associated with the asset or product. Some common cost categories are:
 - » **Initial Purchase Cost:** This includes the upfront cost of acquiring the assets, such as the purchase price and installation expenses. If we exclude the markup we pay for the device, the purchase price consists of the following costs:
 - » **Hardware costs:** This is the total cost to procure materials to build the asset or device as per customer requirements.
 - » **Software costs:** This is the total cost of the software that needs to be developed and installed on the device.
 - » **Deployment costs include** customization or configuration fees for deploying the devices with the necessary settings and applications.

- » **Operating costs:** These include ongoing expenses incurred during the helpful lifetime of the asset, such as maintenance, repairs, upgrades, consumables, and utilities.
 - » **Training costs:** The costs associated with training the personnel to use and maintain the assets effectively.
 - » **Support costs:** Service or warranty costs provided by the supplier.
 - » **Downtime costs:** This estimates potential losses due to the asset's failure or unavailability. This can be costs incurred due to lost productivity, customer dissatisfaction, or reputation damage.
 - » **Residual or disposal value:** The asset's estimated value at the end of its useful life. It can be a positive or negative number depending on whether one can sell/reuse the asset or if one needs to pay for the disposal/recycling of the asset.
- Estimate the helpful lifetime of the asset or device. This may vary depending on the type of device and how it is used.
 - Gather data and quantify each category's costs using primary and secondary sources like historical data, market research, supplier quotes, industry benchmarks, and expert opinions. We may need to discount some costs to factor in inflation and depreciation or amortization, depending on the type of asset.
 - Calculate the present value of all the costs over the useful lifetime of the asset or device and subtract the residual value to calculate the total cost of ownership, i.e.,

$$\text{TCO} = \text{Initial Purchase cost} + \text{Deployment costs} + (\text{Operating costs} + \text{Training costs} + \text{Support costs} + \text{Downtime costs}) * \text{Estimated useful lifetime} - \text{Residual value}$$



The recurring or ongoing expenses like operating costs, training costs, support costs, and downtime costs are the major contributors to the overall TCO, as these are computed over the lifetime of the device, which is long in the case of an enterprise and can sometimes extend as much as 10–15 years.

TCO can vary significantly depending on the type of device, the manufacturer, and the specific needs of an enterprise. For example, a rugged device designed for harsh environments will have a higher TCO than a standard consumer device intended for office use.

Comparing TCO between a GMS and non-GMS Enterprise device

The table below represents an analysis of TCO cost categories between a GMS-enabled enterprise device and a non-GMS Android device running on pure AOSP without any Google applications or services.

TCO cost categories	GMS	AOSP	Reasoning
Initial purchase Cost	↑	↓	As the device manufacturers need to license the GMS bundle for each device, they usually pass on some of the cost to the end customer.
Deployment costs	↓	↑	With various deployment methods, a GMS-enabled enterprise device can be remotely configured, updated, and troubleshooted, thereby reducing the number of field technicians.
Operating costs	↓	↑	Google provides timely software updates and security patches for GMS-enabled devices that are pushed by the device manufacturers within a 90-day window. This reduces outages and the need for unplanned maintenance.
Training costs	→	→	This cost is the same for both types of devices, although Google provides helpful manuals that can be referred to by GMS device customers for training personnel.
Support costs	↓	↑	Part of the service support and automation is already provided by Google for GMS-enabled devices; hence, device manufacturers can reduce the annual support contract fees and provide long warranties at lower costs.
Downtime costs	↓	↑	This is one of the major advantages of a GMS-enabled device, as it reduces downtime risks for end customers through timely software updates and security patches.
Residual value	→	→	The residual value of the hardware is the same for both types of devices, assuming outdated software costs to be nearly zero.

In addition to direct cost savings, GMS-enabled enterprise devices can also help improve productivity and efficiency, leading to indirect cost savings. For example, GMS can provide employees with the tools and resources to work productively from anywhere. GMS-enabled devices can also extend the lifecycles of devices and reduce the need to purchase new devices as often.

Overall, as can be inferred from the above table, GMS-enabled enterprise devices can help organizations reduce their TCO by automating tasks, reducing support costs, extending device lifecycles, and improving productivity.



Conclusion

The incorporation of GMS into enterprise devices demonstrates a potential for enhanced productivity, improved collaboration, and increased user satisfaction. By establishing a comprehensive framework for TCO calculation, this whitepaper facilitates a thorough comparative analysis between AOSP and GMS devices, taking into account various costs throughout the device's lifecycle. This knowledge equips decision-makers with valuable insights to make informed choices that align with their enterprise's priorities and objectives.

About Authors

Senthil Kumar Subramani

Senthil Kumar Subramani is the Senior Manager for Devices Portfolio at Sasken and leads the pre-sales team. He has significant experience in wireless (3G, 4G, and 5G NR) and cloud technologies and has held leadership roles in pre- and post-sales, engineering, and product management. Senthil started his wireless career at Sasken, where he worked on 3G Access Stratum Protocol stack development and collaborated with NXP, France, in delivering solutions to Samsung. Then he worked at Intel Mobile, Munich and then spent nearly eight years at Qualcomm UK in the Customer Engineering division, commercializing Snapdragon chipsets for Tier-1 OEMs in Europe. Later, he gained experience in Radio Access Network (RAN), small cells, and Android while working at Sony Mobile in Sweden, Airspan Communications, Three UK.

Debashis Panigrahi

Debashis Panigrahi has accumulated over 15 years of expertise in product conceptualization and development. He has played a pivotal role in driving Sasken's initiatives in the Android Automotive domain. His involvement dates to the creation of the world's inaugural Android-based smart TV in 2012. Furthermore, he contributed significantly to the development of the world's first Android-based In-Vehicle Infotainment (IVI) system in 2014, during a time when Android was relatively nascent in the automotive sector. He was instrumental in one of the world's first Android-based RSE developments way back in 2018 (for the European market). Recently, he has been leading Sasken efforts on the Android Automotive OS, software-defined vehicles, and enabling edge-AI use cases.



Software Strategist, Tier-1 North American OEM

I want to congratulate you and your team on meeting the milestones planned for the latest Android version upgrade project. You and your team did a great job keeping the project on track to completion despite changing business requirements and priorities during execution. Considering various initial schedule dependency delays that blocked progress, it was truly an amazing job to get to the end of all CTS failures and achieve an acceptable level of monkey test stability in record time. Keep up your good work and continuous support so we can push the SW upgrade to planned customers ASAP!



About Sasken

Sasken is a Product Engineering and Digital Transformation pioneer, delivering concept-to-market and chip-to-cognition R&D solutions to customers across the semiconductor, automotive, industrial, consumer electronics, enterprise devices, satellite communications, telecom, and transportation domains. Sasken has been offering services to global OEMs and Tier-1 in Digital Platform Engineering, AI/ML & Data Analytics, Automation and Cloud Infrastructure Services using Google Cloud Platform. Through the last three decades, Sasken's deep engineering capabilities and technology patents have helped transform over a hundred Fortune 500 clients and powered over a billion digital devices from its state-of-the-art facilities in India, Finland, Germany, the USA, and Japan.

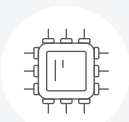
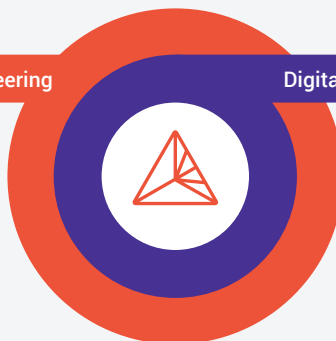
Sasken, with its Android Center of Excellence, has profound lifecycle management experience (New Product Introduction, Operating System Version Upgrade, Security Maintenance Release), enabling our customers' wide range of devices, from smartphones to enterprise/rugged phones and PoS devices. Combining forces with the world's leading mobile operating system vendor, Sasken developed an automated test tool for measuring the device performance of Android Go devices. Additionally, Sasken has a wealth of experience in Android Automotive and Android TV, providing our valued clients with expertise in these domains. This extensive experience is the cornerstone of our innovative approach, enabling us to establish a streamlined factory model. Sasken is dedicated to offering its premium clients highly efficient and remarkably cost-effective solutions through this model.

Sasken is a Android Enterprise member partner and has more than 100 Android Enterprise Expert certified engineers.



Product Engineering

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Comms &
Devices



Automotive



Modem
& Networks



SatCom



Enterprise
Grade Devices



Industrials



Transportation



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